CALENDARS

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↳ The purpose of calendars concept is to find out the day on a particular date of month of the particular year, same calendar year & how many week days in particular months & on which dates etc....

↳ To find out all the above kind of questions we must know the following information.

1) Year / 2 types \ Leap Year → 366 days → 2 odd days
Non-Leap Year → 365 days. → 1 odd day.

↳ Year :- Time taken by earth to revolve around the sun once i.e., approximately $365\frac{1}{4}$ days.

→ But to make fixed no. of days in a year 4 quarters days added to every 4th year & is called leap year.

2) Whether the given year is leap / Not ?

(i) Condition 1:- If the given year is non-century year it must divisibly by 4 or 4 multiple if would be.

(ii) Condition 2:- If the given year is century year.

then it must divisible by 400, otherwise Non-Leap year.

3) Odd days:- After converting the given days into weeks, the remaining days are called odd days.

→ Odd days varies from 0 to 6.

Odd days table:-

Odd day	0	1	2	3	4	5	6
Week day	Sun	Mon	Tue	Wed	Thu	Fri	Sat

4) Odd days in Leap & Non-Leap years:-

(i) Odd days in Leap:- $\frac{366 \text{ days}}{7}$ = 52 weeks 2 days.

∴ 2 odd days.

(ii) Odd days in Non-Leap:- $\frac{365 \text{ days}}{7}$ = 52 weeks 1 day

∴ 1 odd day.

5) (i) To find odd days ⇒ $\frac{\text{Given days}}{7}$ ⇒ take Remainder

(ii) To find no. of leap years ⇒ $\frac{\text{years}}{4}$ ⇒ take Quotient.

6) Odd days in centuries:-

(i) 100 years:- No. of leap years in 100 yrs ⇒ $\frac{100 \text{ yr}}{4}$ = 25 yrs

⇒ 100th year century is not divisible by 400 so it is to be subtracted from 25. i.e., 24 leap years & each leap year is having 2 odd days.

(158) So, 100

24 L.Y (100-24) = 76 N.L.Y

24×2 + 76×1 = 124 odd days.

odd days = $\frac{124}{7}$

= 17 weeks 5 odd days

⇒ 100 years ⇒ [5] odd days

2) 200 years :- 200th year also non-leap year. So, first 100 years & 2nd 100 yrs are of same odd days.

↳ 200 yrs = 100 + 100 = 5 odd + 5 odd = $\frac{10}{7}$ ⇒ [3] odd days

3) 300 yrs :- 300th year also non-leap year. 1st, 2nd & 3rd centuries are having same no. of odd days.

↳ 300 yrs = 100+100+100 = 5+5+5 = 15 odd = $\frac{15}{7}$ ⇒ [1] odd day

4) 400 yrs :- 400th year is leap year & it's having 2 odd days means 4th century is having one extra day i.e.,

↳ 400 yrs = 100+100+100+100 = 5+5+5+[5+1] = 21 odd = $\frac{21}{7}$ ⇒ [0] odd days

5) 500 yrs :- 100 yrs from 401 to 500 also like 1st 100 yrs because 500th yr also non-leap year.

500 yrs = 400+100 = 0+5 = 5 odd days.

↳ Means odd days for centuries are cyclic as 5, 3, 1 & 0. for every 400 yrs.

(7) If 1st day of non-leap year is any day of week then it's next year 1st day is +1 day because it is having one odd day.

(ii) If non-leap year 1st day is any day of week. means it's last day also same day of week.

(iii) If leap year 1st day is any day of week then it's next year 1st day is +2nd day because it's having 2 odd days.

(iv) If leap year 1st day is any day of week then its last day is '+1' day.

(8) Odd days table for centuries:-

Odd days	~~first~~ ~~last~~ Centuries					first	Last
5	1000	500	900	1300	1700	Fri	Fri
3	2000	600	1000	1400	1800	Wed	Wed
1	300	700	1100	1500	1900	Mon	Mon
0	400	800	1200	1600	2000	Sat	Sun

↳ Odd day means last day of that particular given period.

(9) (i) First day of any century should be either Fri/Wed/Mon/Sat. & should not be Sun/Tue/Thur.

(ii) Last day of any century should be either Fri/Wed/Mon/Sun & should not be Tue/Thur/Sat.

(10) Maximum gap b/w 2 leap years:- Generally gap b/w 2 leap yr's is 4 yr's but when non-leap century year comes b/w 2 leap years then maximum gap